Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027(ODD)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Srushti Chaudhari



16-08-2026
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**LAB: Software Engineering**
Semester	**7th**	**Course Code**	**23CT1704**
Roll No	**24**	**Name of Student**	**Srushti Chaudhari**

Practical Number	**1**
Course Outcome	Upon successful completion of the course the students will be able to; 1. Choose appropriate software engineering process model, requirement engineering principles and software designing fundamentals for a given project. 2. Select appropriate testing strategy and apply testing principles for testing a given application. 3. Apply basics of software configuration management, version control and change control in software development. 4. Evaluate cost estimation, effort and severity of software risk for given application. 5. Perform basic operations on Sub-version for software version control.
Aim	Introduction to Software Engineering fundamentals, UML and Star UML
Problem Definition	To study the fundamental concepts of Software Engineering, understand the Unified Modeling Language (UML), and learn to create and represent software system models using StarUML. The practical focuses on understanding different UML diagrams and their purpose in software development.
Theory (100 words)	1.Introduction to Software Engineering & UML Introduction to Software Engineering Software Engineering is a systematic, disciplined and quantifiable approach to the design, development, testing, and maintenance of software. It applies engineering principles so that the final product is reliable, efficient, scalable, and meets the user's requirements within time and budget. Why Software Engineering is needed: • Software has become large and complex — an unplanned approach leads to failure. • It ensures the product is delivered on time and within budget. • It improves software quality, reliability and maintainability. • It provides a structured way to manage changing requirements.

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2. Software Development Life Cycle (SDLC)
The SDLC is a step-by-step process followed to design, develop, and test high-quality software. Each phase produces an output that becomes the input for the next phase.
1. Requirement Analysis Gathering and documenting functional & non-functional requirements. Output: SRS document.
2. System Design Creating the architecture, database design and UI design (using UML diagrams).
3. Implementation (Coding) Writing the actual source code based on the design.
4. Testing Verifying the software is defect-free (unit, integration, system, acceptance testing).
5. Deployment & Maintenance Releasing the software and providing ongoing support/bug fixes.

What is UML?
UML (Unified Modeling Language) is a standardized, general-purpose visual modeling language used in object-oriented software engineering to specify, visualize, construct, and document a software system. It was developed by Grady Booch, Ivar Jacobson and James Rumbaugh, and is now maintained by the OMG. UML is not a programming language it is a diagrammatic notation.

Advantages of UML:
• Provides a standard, easy-to-understand visual notation.
• Improves understanding of the system through graphical representation.
• Facilitates communication between developers, clients and stakeholders.
• Supports the entire SDLC, from analysis to implementation.
• Independent of any specific programming language.
• Reduces development cost and time by catching design flaws early.

Types of UML Diagrams
UML diagrams are broadly classified into two categories: Structural (Static) diagrams and Behavioral (Dynamic) diagrams.

A. Structural Diagrams
   • Class Diagram — Shows classes, attributes, operations and relationships.
   • Object Diagram — Represents a snapshot of object instances at a moment in time.
   • Component Diagram — Shows how physical components/modules are organized.
   • Package Diagram — Groups related elements into packages.
   • Deployment Diagram — Shows the physical/hardware setup and deployment.

B. Behavioral Diagrams
   • Use Case Diagram — Shows functional requirements via actors and use cases.
   • Sequence Diagram — Shows the order of messages exchanged between objects over time.
   • Activity Diagram — Represents workflow of activities and decisions (like a flowchart).
   • State Diagram — Shows the states an object passes through during its lifetime.
   • Communication Diagram — Shows interaction between objects, focused on structural links.

Important UML Diagrams

1. Use Case Diagram
   Example — Library Management System:
   • Actors: Member, Librarian

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• Use Cases: Search Book, Borrow Book, Return Book, Manage Catalog (Actors are shown as stick figures; use cases as ovals inside a system-boundary rectangle; lines connect actors to the use cases they perform.)

2. Class Diagram
Example — Library Management System:
• Book (- title, - isbn / + getDetails())
• Library (- name / + addBook())
• Member (- name, - id / + register()) (Each class box has 3 parts: name, attributes, operations. Library — Book is an aggregation; Library — Member is an association.)

Relationships Between Two Classes

Relationship	Meaning	Exam
Inheritance	IS-A relationship; child inherits from parent. Solid line + hollow triangle.	Car IS
Association	General structural link between two classes. Simple solid line.	Stude
Aggregation	Weak HAS-A; part can exist independently. Solid line + hollow diamond.	Depar Profes
Composition	Strong HAS-A; part cannot exist without whole. Solid line + filled diamond.	House
Dependency	One class temporarily uses another. Dashed line + open arrow.	Order Paym

3. Sequence Diagram
Example — Library Management System (Borrow Book):
• Objects: Member, Library, Book
• Messages: searchBook(), checkAvailability(), available (return), issueBook(), updateCatalog() (Objects are shown as boxes with vertical lifelines; messages are horizontal arrows in time order from top to bottom.)

4. Activity Diagram
Example — Library Management System (Borrow Book):
• Start → Search for book → Decision: Available?
• Yes → Issue book → Update catalog → End
• No → Show "Not Available" message → End (Activities are rounded rectangles, decisions are diamonds, start/end are filled circles.)

5. State Diagram
Example — Order object (E-commerce system):
• States: New, Processing, Confirmed, Cancelled, Complete
• Transitions: submit() → Processing, paymentReceived() → Confirmed, cancel() → Cancelled, ship() → Complete (States are rounded rectangles, transitions are labeled arrows, start/end are filled circles.)

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UML Symbols

Symbol	Meaning
Rectangle	Class (also object / system bound
Oval	Use Case
Stick Figure	Actor
Arrow (hollow triangle)	Inheritance
Arrow (dashed, open head)	Dependency
Plain Line	Association / Relationship
Diamond (filled)	Composition
Diamond (hollow)	Aggregation
Rounded Rectangle	Activity / State

Procedure and Execution

(100 Words)

Algorithm(Steps):

1. Start.
2. Open StarUML.
3. Create a new UML project.
4. Select the required UML diagram.
5. Identify the entities/components required for the system.
6. Add the required UML elements to the diagram.
7. Define appropriate relationships between the elements.
8. Enter names and properties of the elements.
9. Arrange and validate the diagram.
10. Save the UML project.
11. Generate/export the diagram.
12. Stop.

Code:

Not applicable for this practical because UML diagrams are graphical models rather than executable programs.

The completed UML diagram created in StarUML should be inserted in this section if required by the practical format.

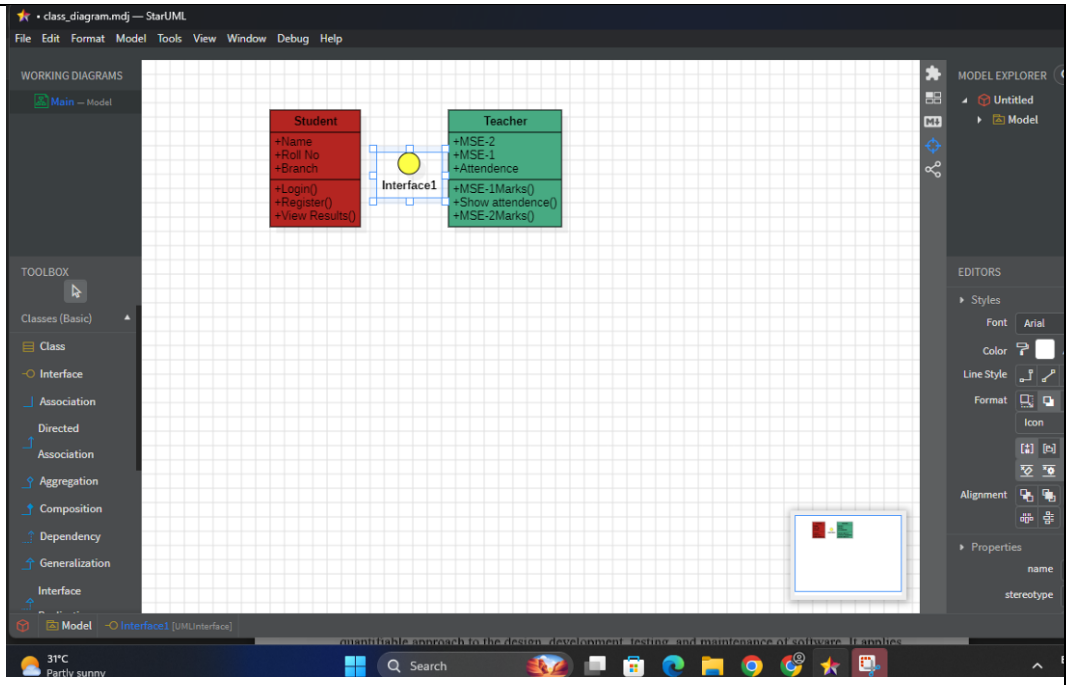
Output:

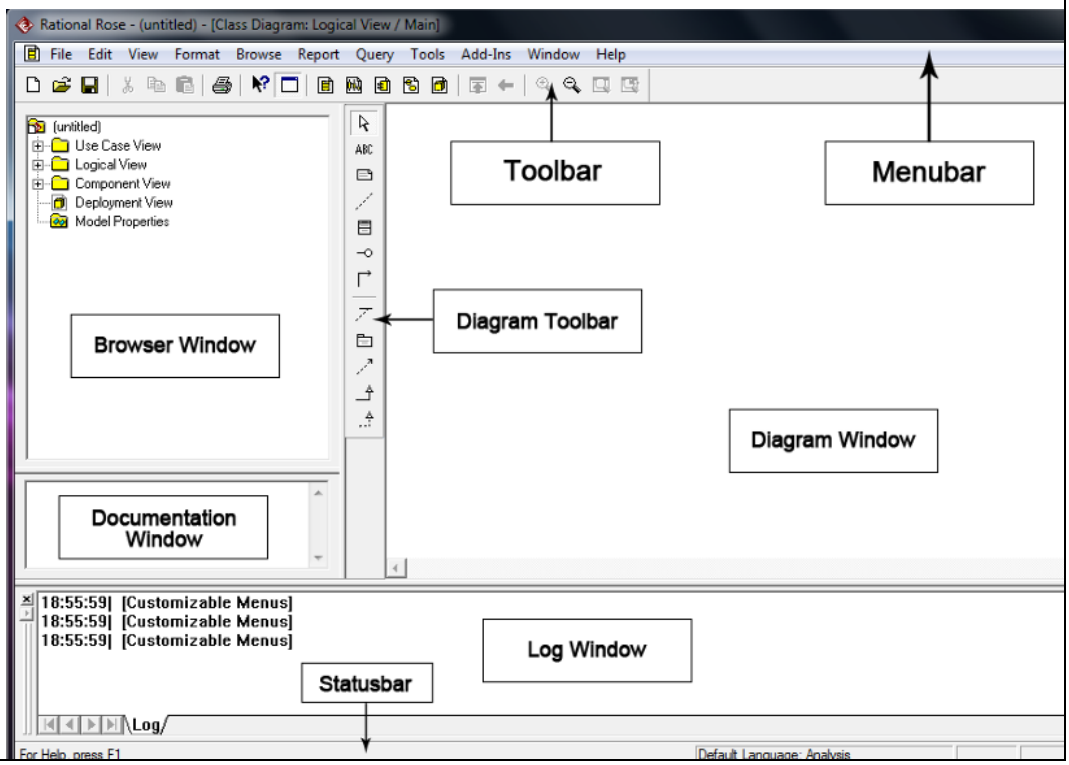
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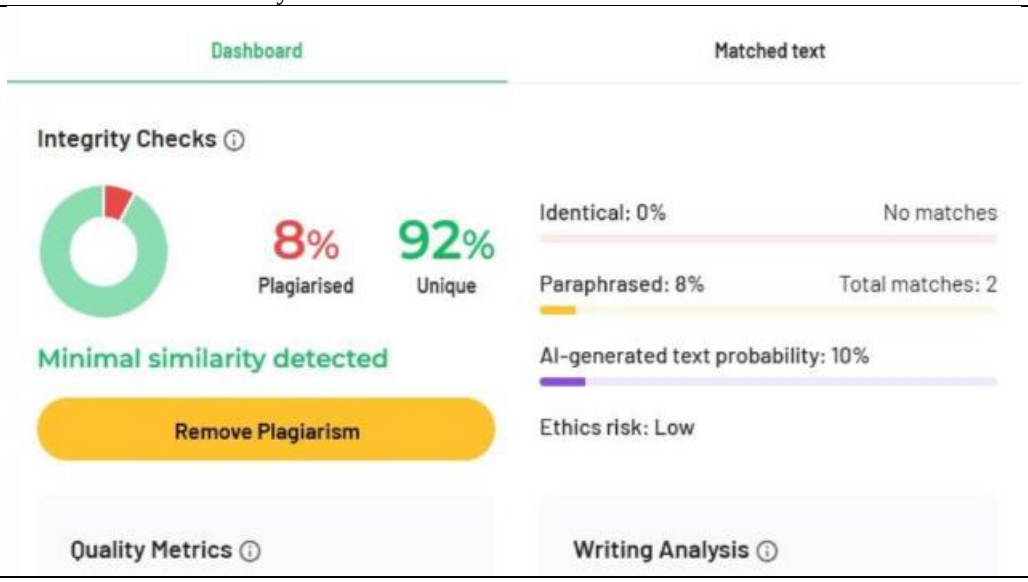


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Output Analysis	Introduction to Rational Rose Rational Rose is a visual modeling (CASE) tool developed by Rational Software Corporation (later acquired by IBM) that is used to design, develop, and document software systems using the Unified Modeling Language (UML). It's widely used in academic software engineering labs to draw UML diagrams instead of drawing them by hand. Features: • Drag-and-drop diagram creation (actors, classes, use cases, etc.) • Round-trip engineering — can generate code (C++, Java, Visual Basic, etc.) from UML diagrams, and reverse-engineer existing code back into UML diagrams. • Supports multiple diagram views of the same system, all linked to a single underlying model, so changes stay consistent. • Document generation for reports and specifications. Advantages: • Speeds up the design phase and reduces manual drawing errors. • Improves consistency between diagrams (since they share one model). • Makes it easier to visualize and communicate system architecture to a team or client. • Bridges the gap between design and implementation via code generation.
Link of student Github profile where lab assignment has been uploaded	https://github.com/SRchaudhari15/Software-Engineering-Practicals-24.git
Conclusion	We have studied the basics of Software Engineering, the phases of SDLC, and the Unified Modeling Language (UML) including its definition, advantages, classification of diagrams, important diagrams with examples, relationships between classes, and standard UML symbols used to model a software system.
Plag Report (Similarity index < 12%)	

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